

1. Take a string input and print it in uppercase.
2. Take a string input and print it in lowercase.
3. Count the total number of characters in a string.
4. Print the first and last character of a string.
5. Reverse a string.

Example:

Input: hello

Output: olleh

6. Count vowels in a string.

Example:

Input: education

Output: 5

7. Count consonants in a string.
8. Check whether a character exists in a string or not.
9. Replace all spaces with _.
10. Count how many times a particular character appears.

Example:

Input: banana

Character: a

Output: 3

Easy-Intermediate Level

11. Check whether a string is palindrome or not.

Example:

madam → Palindrome

hello → Not Palindrome

12. Count words in a sentence.
13. Find the longest word in a sentence.
14. Remove all vowels from a string.

15. Print all characters one by one using loop.

16. Convert first letter of every word into uppercase.

Example:

hello world
Hello World

17. Find duplicate characters in a string.

Example:

programming
r, g, m

18. Check whether two strings are anagrams.

Example:

listen
silent

19. Remove duplicate characters from string.

Example:

programming
progamin

20. Count digits, alphabets, and special characters separately.

Intermediate Level

21. Find frequency of each character.

Example:

apple
a = 1
p = 2
l = 1
e = 1

22. Find the second most repeated character.

23. Compress a string.

Example:

aaabbcccd
a3b2c4d1

24. Expand compressed string.

Example:

a3b2c1
aaabbc

25. Remove all extra spaces from sentence.

Example:

"hello world"
"hello world"

26. Find all substrings of a string.

27. Find all permutations of a string.

28. Check if two strings are rotations of each other.

Example:

ABCD
CDAB

29. Find largest and smallest word in a sentence.

30. Sort words alphabetically.

Advanced Level

31. Find the first non-repeating character.

Example:

aabbcddee
c

32. Find the first repeating character.

33. Implement your own split() function without using split.

34. Implement your own replace() function.

35. Check whether a password is strong.

Conditions:

- At least 8 characters
- One uppercase
- One lowercase
- One digit
- One special character

36. Make a word guessing game.

37. Create a username generator from full name.

Example:

Kana Kumari

kana123

38. Encrypt a string using Caesar Cipher.

39. Decrypt Caesar Cipher.

40. Create a mini chatbot using string matching.

Loop + String Pattern Problems

41.

Input: PYTHON

Output:

P

PY

PYT

PYTH

42.

Input: PYTHON

Output:

PYTHON

PYTHO

PYTH

43.

Input: ABCDE

Output:

A

BB

CCC

DDDD

44. Print characters with index numbers.

Example:

0 -> H

1 -> e

45. Separate uppercase and lowercase characters.

Real Practice Challenge

46. Email validator

47. Phone number validator

48. Count emojis/special symbols in text

49. Find most common word in paragraph

50. Create a text-based login system using strings